

Training V8 Report: Shot Classifier (Full Skeleton)

Overview

This report documents the results of Phase 3, Version 8. Following the failure of V5 (33% accuracy due to "Feature Starvation"), we implemented the **V7/V8 Strategy**: using the full 99-feature skeleton (33 landmarks × 3 dimensions) instead of just 7 angles.

Objective: Train a "Base Model" (Pro Cricketer Brain) on 1,517 videos to recognize 4 distinct cricket shots.

Dataset Summary

- **Dataset:** Combined Kaggle + Custom (Cleaned & Filtered)
- **Total Samples:** 1,517 videos
- **Classes (4):** drive, pull, sweep, cut (Lofted was removed to reduce variance)
- **Feature Shape:** (50, 99) — Full 3D Skeleton Trajectory

Model Architecture (V8)

- **Type:** Bidirectional LSTM (Sequence Model)
- **Input:** 50 frames of 99 coordinates.
- **Core Layers:**
 - Bidirectional(LSTM(128)) - Captures detailed motion context.
 - Bidirectional(LSTM(64)) - The "Brain" layer (extracted for Transfer Learning).
 - Dense(64) - Interpretation layer.

Training Performance

- **Optimization:** Used ReduceLROnPlateau to fine-tune the learning rate when progress stalled.
- **Early Stopping:** Stopped at **Epoch 47** (Restored best weights from Epoch 32) to prevent overfitting.
- **Loss:** Decreased steadily from 2.59 to 0.52, indicating healthy learning.

Final Evaluation Results

Overall Accuracy: 70.18%

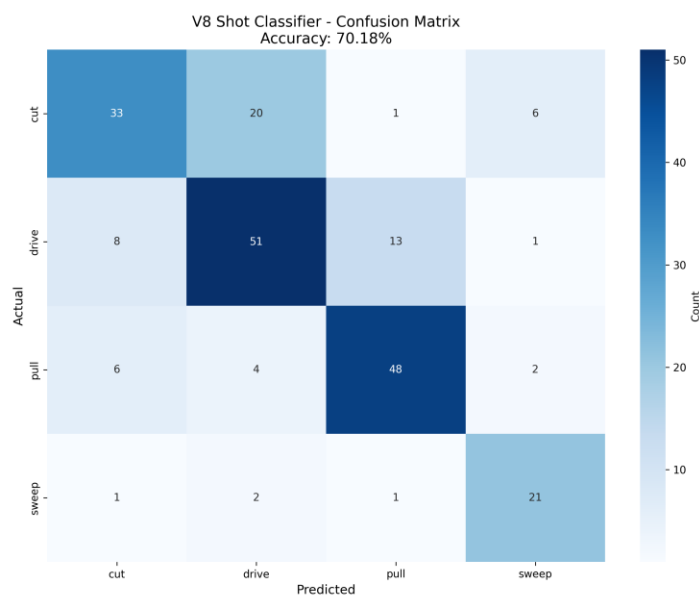
(Baseline Random Guessing: 25.00%)

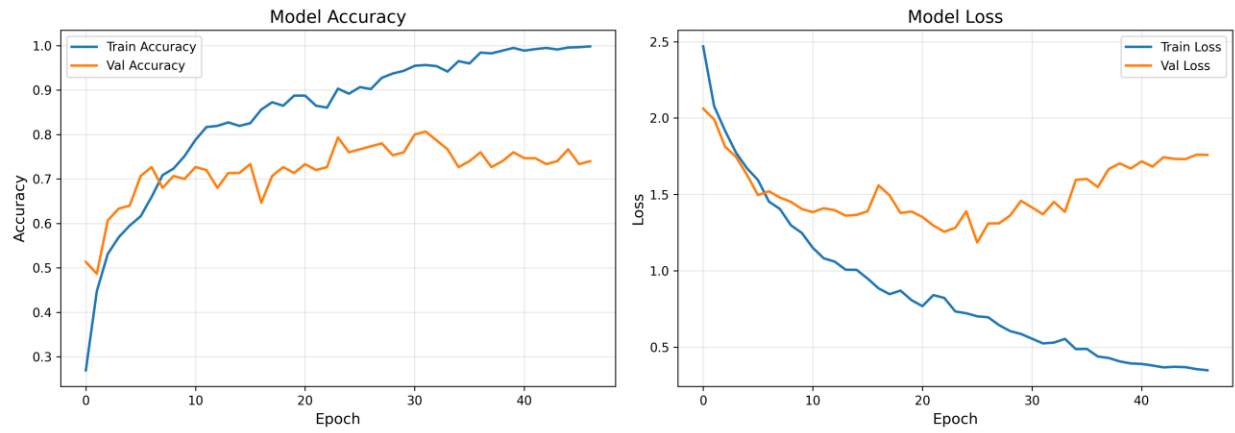
Per-Class Performance

Shot Type	Precision	Recall (Accuracy)	F1-Score	Status
Sweep	70%	84%	0.76	🟢 Excellent
Pull	76%	80%	0.78	🟢 Excellent
Drive	66%	70%	0.68	🟡 Good
Cut	69%	55%	0.61	🔴 Weak

Analysis & Conclusion

- Hypothesis Confirmed:** The jump from 33% (V5) to 70% (V8) proves that **Feature Engineering** was the bottleneck. The AI needed the full skeleton to understand the shots.
- Strong Biomechanics:** The model is highly effective at recognizing shots with distinct body shapes (sweep with knee down, pull with cross-bat).
- Confusion Areas:** The cut shot (55%) is the hardest to classify, likely because it biomechanically resembles a drive in the initial phase.
- Ready for Transfer:** The model has learned the "Physics of Cricket" well enough (70% general accuracy) to serve as a "Teacher" for the Error Classifier.





Next Step: Proceed to train **Model 2 (Error Classifier)** using Transfer Learning from this V8 Brain.